

Title

Incidence and Timing of Heel Ulceration Across Acute Care Settings: A Pathway Analysis

Abstract

We prospectively followed 183 hospitalized patients with neurological or functional impairments across pre-hospital transport, emergency care, and hospital wards at two Swedish academic centers. One group (91 patients) received a protective heel device beginning in the ambulance; 92 controls received routine care. Among the entire cohort, 38 individuals (21%) developed heel tissue breakdown at various points in the care pathway. In the device group, 18% developed tissue injury compared to 24% in controls. The majority of pressure-related damage occurred during the ward phase (59% of all cases), though early signs were detected in 12% of patients during emergency department evaluation. Univariate analysis identified admission severity score, duration of bed rest, and delayed mobility as predictive of tissue breakdown ($p < 0.05$). Skin inspection protocols alone did not prevent most injuries. Device use was associated with reduced injury progression to deeper tissue layers. These results suggest that multi-phase prevention strategies beginning in prehospital care, combined with systematic assessment at ward admission, may improve outcomes in this vulnerable population.